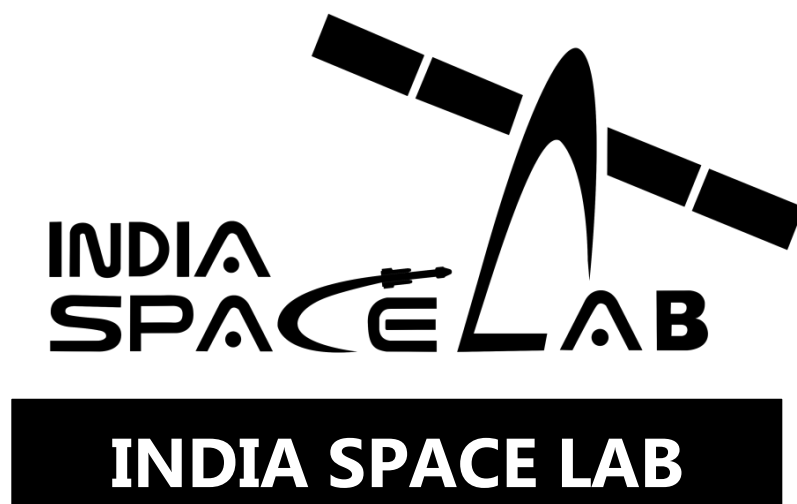


MODEL ROCKETRY
ROCKETRY TRAINING – AEROSPACE SIMULATION
WORKSHOP PROJECT



Project Report
01ST May to 15th June 2026

Student Name: GANGINENI RUSHIKESAV
Institute Name: JNTUA College of Engineering
Ananthapuramu
Institute Roll no: 23001A0451
Enrollment no: ISL-878154

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1. AIM

The objective of this project is to perform a comprehensive computational analysis on a passive stabilization rocket fin assembly to evaluate its structural safety and aerodynamic efficiency. This involves:

1. **Finite Element Method (FEM):** Executing a static structural simulation to determine stress distribution, deformation, and the structural Factor of Safety (FoS) under severe aerodynamic pressure loads.
2. **Computational Fluid Dynamics (CFD):** Executing a steady-state fluid flow simulation to visualize the aerodynamic wake, pressure contours, and extract the total aerodynamic drag force during a 50 m/s ascent phase.

2. THEORY (PART A - THEORETICAL SECTION)

Q1. Explain Finite Element Method (FEM) in your own words.

The Finite Element Method (FEM) is a numerical mathematical technique used to solve complex physics and engineering problems. It works by taking a complex, continuous 3D solid geometry and breaking it down (discretizing it) into thousands or millions of smaller, simpler geometric shapes called "finite elements" (such as tetrahedrons). The computer solves physical equations at the nodes connecting these elements. In aerospace engineering, FEM is heavily used to simulate how rocket components will react to physical stresses, extreme thermal gradients, and aerodynamic vibrations without having to build and break expensive physical prototypes.

Q2. What is Von Mises Stress and why is it important?

Von Mises Stress is a theoretical equivalent stress value used to predict the yielding of ductile materials (like the aluminum used in rockets) when subjected to complex, 3D multi-directional forces. It combines all principal stresses (X, Y, and Z axes) into a single scalar value. It is important because engineers compare this calculated Von Mises Stress directly against the material's known "Yield Strength." If the Von Mises stress exceeds the yield strength, the material will undergo permanent, plastic deformation—constituting a structural failure.

Q3. Explain the role of meshing in FEM simulations.

Meshing is the foundation of FEM. It is the physical act of dividing the CAD model into the grid of finite elements. The accuracy of a simulation is highly dependent on mesh quality. A "coarse" mesh (few elements) computes very quickly but produces highly inaccurate, blocky results. A "fine" mesh (many elements) produces highly accurate results but carries a massive computational cost in time and memory. "Mesh dependency" occurs when the answers change simply because the grid size changed. To prove a result is mathematically true, engineers must perform a mesh convergence study until making the mesh finer no longer significantly changes the result.

Q4. Differentiate between CFD and FEM.

- **FEM (Computational Solid Mechanics):** Focuses on solid objects. Its objective is to determine how structures bend, break, or heat up under loads. Outputs include Von Mises Stress, Strain, Displacement, and Factor of Safety.
- **CFD (Computational Fluid Dynamics):** Focuses on fluids (gases and liquids). Its objective is to determine how air or liquid flows around or through an object by solving the Navier-Stokes equations. Outputs include Velocity, Pressure, Drag Force, Lift, and boundary layer separation.

Q5. Why are boundary conditions important in simulations?

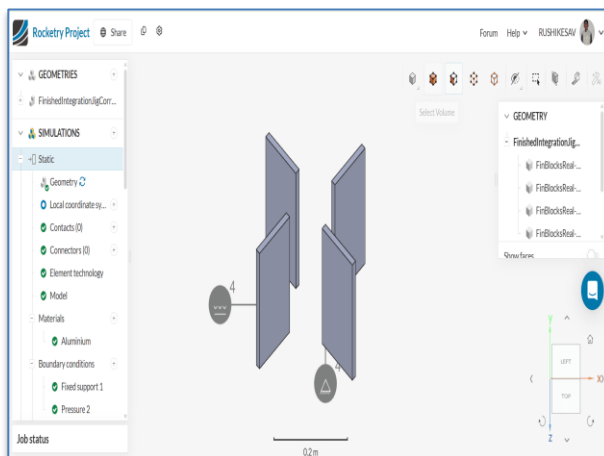
Boundary conditions (BCs) are the mathematical rules applied to the simulation to connect the virtual model to real-world physics. They tell the computer exactly how the object is held in place (Supports) and what forces are acting upon it (Loads). If incorrect BCs are assumed (e.g., assuming a fin is glued on instead of bolted, or assuming an inlet velocity of 5 m/s instead of 50 m/s), the computer will solve the wrong physics problem perfectly. Accurate boundary conditions are the absolute key to obtaining useful engineering data.

3. GEOMETRY

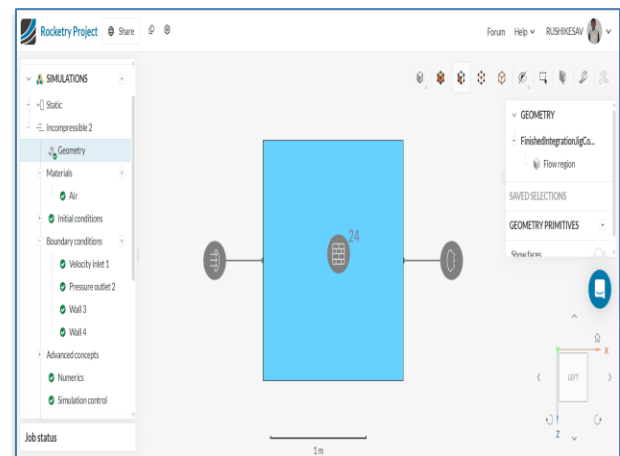
Task 1: Geometry Creation

The geometry utilized for this project is a four-fin passive stabilization suborbital rocket assembly.

- **Solid Geometry (FEM):** The baseline CAD model consisted of four discrete fin bodies integrated onto a central axis.
- **Flow Domain (CFD):** An External Flow Volume (a virtual wind tunnel) was generated as a bounding box around the solid geometry. The box was extended significantly upstream and downstream of the fins to ensure that the artificial domain walls did not compress the aerodynamic wake or artificially accelerate the flow.



Screenshot of the 3D CAD model showing the fins



Screenshot of the transparent blue "Flow Region"

bounding box created around the fins

4. MATERIAL PROPERTIES

To accurately capture the physics of the assembly, the following material assignments were established:

Solid Mechanics (Rocket Fins): Aerospace-Grade Aluminum 6061-T6 (Isotropic Linear Elastic)

- Density (ρ): 2,700 kg/m³
- Young's Modulus (E): 68.9 GPa
- Poisson's Ratio (ν): 0.33
- Tensile Yield Strength: 276 MPa

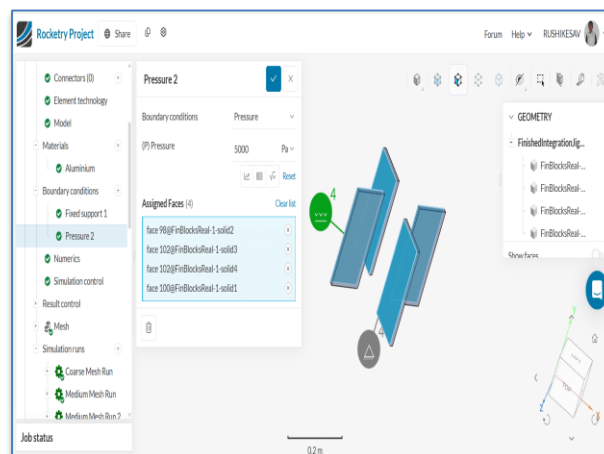
Fluid Dynamics (Wind Tunnel): Standard Air (Incompressible)

- Density (ρ): 1.196 kg/m³
- Kinematic Viscosity (ν): 1.529×10^{-5} m²/s

5. BOUNDARY CONDITIONS

FEM Boundary Conditions:

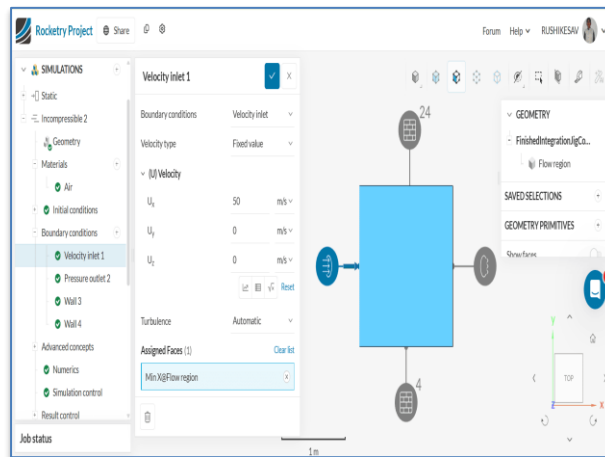
- **Fixed Supports:** Applied to the root faces of the fins where they connect to the airframe, mathematically locking all degrees of freedom (0 translation, 0 rotation).
- **Aerodynamic Loads:** A uniform static pressure load of **5,000 Pa** was applied normally to the lateral faces of the fins to simulate a worst-case scenario, high-Angle-of-Attack (AoA) lateral wind gust during Maximum Dynamic Pressure (Max-Q).



Screenshot of the FEM Boundary conditions setup showing the red/blue arrows pointing at the fins

CFD Boundary Conditions:

- **Velocity Inlet:** 50 m/s (Applied to the front face of the flow domain).
- **Pressure Outlet:** 0 Pa Gauge (Applied to the rear face, allowing free exhaust).
- **Outer Domain Walls:** Slip Walls (Zero friction on the tunnel perimeter).
- **Fin Surfaces:** No-Slip Walls (Applied to the 24 internal hole faces representing the fins to calculate aerodynamic skin friction).



Screenshot of the CFD Velocity Inlet or Pressure Outlet setup

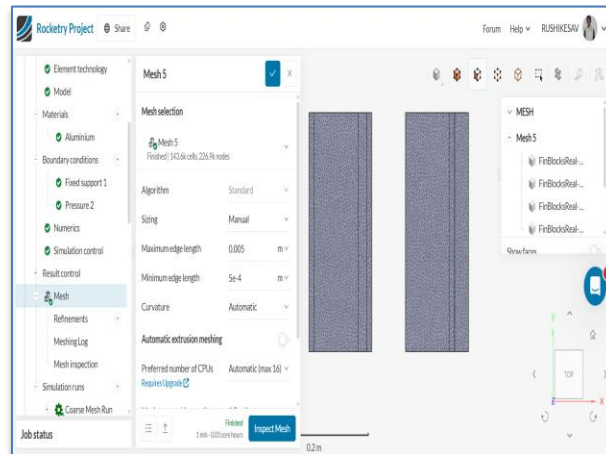
6. MESH DETAILS

FEM Mesh Study (Convergence):

A rigorous mesh convergence study was performed to ensure grid independence.

1. **Coarse Mesh:** 682 cells $\rightarrow$ Max Displacement: 0.00787 mm
2. **Medium Mesh:** 18,200 cells $\rightarrow$ Max Displacement: 0.00829 mm
3. **Fine Mesh:** 143,600 cells $\rightarrow$ Max Displacement: 0.00836 mm

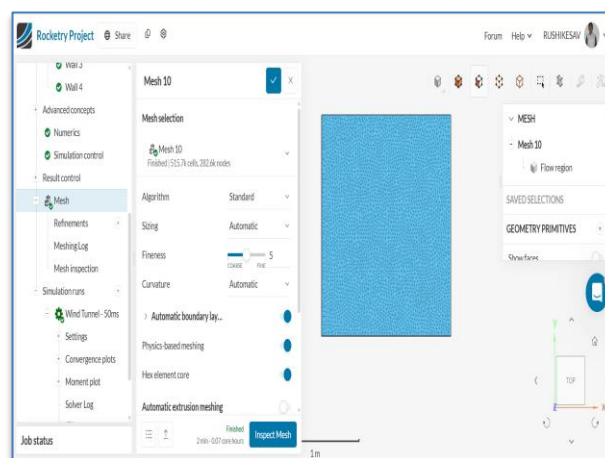
The variance between the Medium and Fine mesh displacements dropped to less than 1%, mathematically proving convergence. The **143.6k Fine Mesh** was utilized for final FEM results.



Screenshot of the structural mesh applied to the fins

CFD Mesh Strategy:

The fluid domain utilized an automated hex-dominant meshing algorithm. Due to the complex nature of fluid physics, a much higher resolution was required. The final CFD mesh consisted of **515,700 cells**. The refinement strategy included adding automatic boundary layer inflation layers (prismatic cells) against the fin walls to accurately capture the steep velocity gradients required for the $k - \omega$ SST turbulence model.



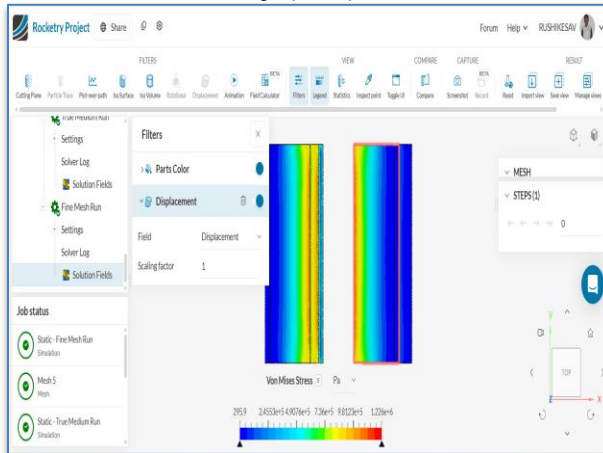
Screenshot of the CFD mesh showing the 515.7k cell count

7. RESULTS

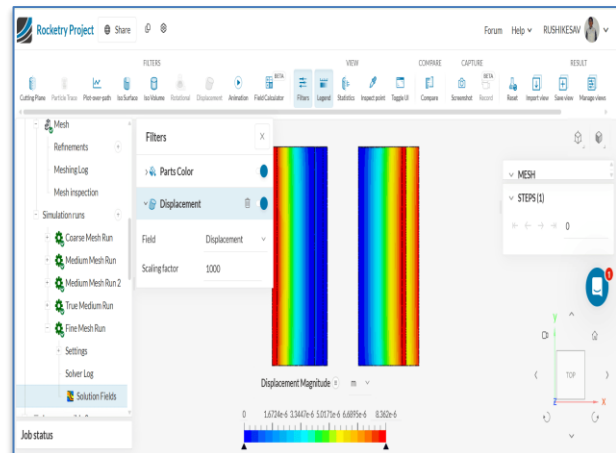
7.1 FEM Results (Structural Integrity)

The linear static analysis under a 5,000 Pa load yielded the following maximum values:

- **Maximum Von Mises Stress:** 1.226 MPa
- **Maximum Deformation:** 0.00836 mm
- **Factor of Safety (FoS):** 225



Screenshot of the Von Mises Stress contour plot

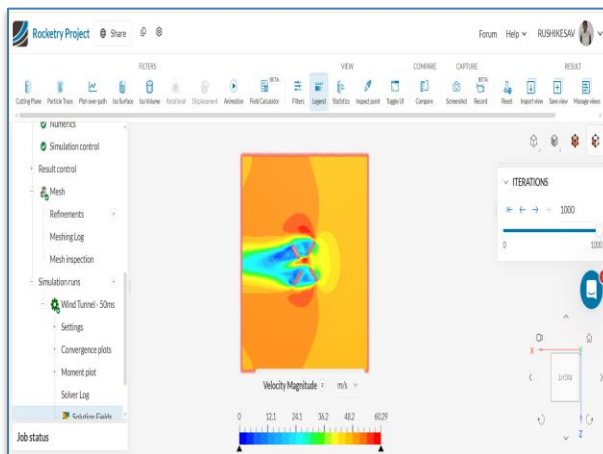


Screenshot of the structural Deformation plot

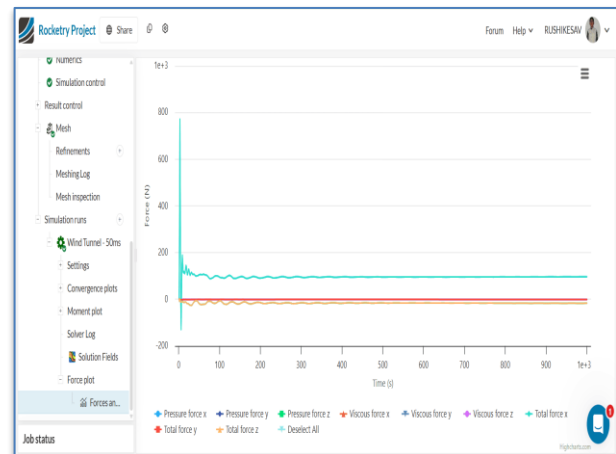
7.2 CFD Results (Aerodynamics)

The steady-state, incompressible flow simulation converged successfully over 1,000 iterations.

- **Aerodynamic Drag:** The steady-state aerodynamic drag force opposing the 50 m/s airflow was measured at exactly **100 N**.



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1.Screenshot of the CFD Cutting Plane showing the Velocity Magnitude colors cutting through the fins

2.Screenshot of the Force Plot graph showing the line flatlining at 100 N

8. DISCUSSION

Result Interpretation and Engineering Reasoning

- **Structural Discussion:** The Von Mises stress contour revealed that the highest stresses were isolated entirely at the root leading edge of the fins—this is the fulcrum where the cantilevered structure bears the highest bending moment. However, because the peak stress (1.226 MPa) is vastly lower than the yield limit of Aluminum 6061-T6 (276 MPa), the structural failure risk is effectively zero. The Factor of Safety is evaluated as $FoS = 276/1.226 = 225$. While highly safe, this indicates the fins are heavily over-engineered and carry unnecessary parasitic mass.
- **Aerodynamic Discussion:** The velocity contour planes provided excellent visualization of the aerodynamic behavior. High-pressure, low-velocity stagnation zones formed exactly at the blunt leading edges. Flow remained cleanly attached along the lateral fin faces, indicating a relatively streamlined design. A distinct, low-velocity turbulent wake formed at the trailing edges, which is the primary contributor to the 100 N of profile drag measured by the force plot. The fact that the force convergence plot flatlined smoothly indicates that the wake is stable and predictable, rather than chaotically fluttering.

9. CONCLUSION & BONUS PROJECT DISCUSSION

The dual-phase computational validation of the suborbital fin assembly was highly successful. The FEM simulation confirms that the structure will easily withstand massive 5000 Pa lateral loads without yielding, proving flight safety. The CFD simulation successfully characterized the aerodynamic profile, resulting in a predictable 100 N of drag at 50 m/s.

Bonus Task Reasoning (Design Optimization):

Based on the engineering reasoning derived from these results, the current original design requires optimization. An FoS of 225 translates to excessive aluminum mass, which severely penalizes the rocket's apogee and Delta-V. The recommended design optimization is to reduce the root thickness of the fins and apply a tapered aerodynamic sweep. By intentionally reducing the material thickness, we will simultaneously lower the total mass, reduce the frontal cross-sectional area (thereby lowering the aerodynamic drag below 100 N), and bring the Factor of Safety down to a more optimal, industry-standard margin of 2.0 to 3.0.

10. REFERENCES

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3. Logan, D. L. A First Course in the Finite Element Method (6th Edition). Cengage Learning, 2016. (Used for Von Mises yield criteria and mesh convergence methodology).